CASE FILE: ALGEBRA

Equations & Inequalities Detective Investigation

ALGEBRA I HONORS - 7TH GRADE

FINAL PRODUCT	Canva Presentation / Digital Project
PROJECT SCORE	100 Points
GRADE WEIGHT	70% of Major Assessment Grade
REQUIRED PARTS	5 + Resources Used

YOUR MISSION

You are an **Algebra Detective**. Investigate a real-world problem where someone must choose between **two options**. You will research accurate information, use equations and inequalities as mathematical evidence, and make a final recommendation.

Which option should be chosen, and what mathematical evidence supports your decision?

FINAL PRODUCT: CANVA

Create your final project in **Canva**. Your design may be a presentation, infographic-style presentation, or another teacher-approved Canva format. It must be easy to read and organized so your teacher can quickly locate all five required parts and your Resources Used page.

RESEARCH & TECHNOLOGY RULES

Use the Internet to research accurate, realistic, and current costs. Do not make up prices when real information is available. Examples include current monthly fees, rental prices, ticket costs, membership fees, per-person charges, mileage rates, or other information connected to your case.

AI MAY NOT BE USED TO COMPLETE THIS PROJECT. Do not use ChatGPT or another AI tool to choose your case, find prices, create equations or inequalities, solve the mathematics, write explanations, create your final verdict, or generate the content/design for your Canva project. The thinking, mathematics, writing, and design choices submitted must be your own.

Keep track of your sources. You must include a **Resources Used** page at the end of your Canva project. List the name of each website or source and the web address. If you use information from a business website, identify the business and the page where you found the price or information.

Tip: Research first. Write down the price, fee, date you found it, and website immediately so you do not have to search for the source again later.

YOUR CANVA PROJECT: 5 REQUIRED PARTS

Use these five headings. You may use more than one Canva page for a part if needed.

PART 1 - CASE FILE: Scenario & Clues

Introduce the investigation.

[ ] Give your case a creative title.
[ ] Describe the real-world situation in your own words.
[ ] Clearly identify **Option A** and **Option B**.
[ ] Use Internet research to find accurate costs or other needed information.
[ ] List the important prices, fees, limits, or other clues.
[ ] Define your variable and include its unit.
[ ] Include at least one realistic restriction such as a budget, minimum, maximum, or range.
[ ] Write your **Central Question**: Which option should be chosen?

PART 2 - EVIDENCE #1: Solving Equations

Use equations to uncover information.

[ ] Create and solve **one multi-step equation** connected to your researched case.
[ ] Create and solve **one equation with variables on both sides**.
[ ] Show every algebra step.
[ ] Check at least one solution by substitution.
[ ] Write a complete sentence explaining what each answer means.
[ ] Your variables-on-both-sides equation may determine when the two options have the same cost or value.

PART 3 - EVIDENCE #2: Solving Inequalities

Use inequalities to find limits and possible solutions.

[ ] Create and solve **one multi-step inequality**.
[ ] Create and solve **one compound inequality**.
[ ] Show every algebra step.
[ ] Reverse the inequality symbol when multiplying or dividing by a negative number.
[ ] **Graph each solution on a number line.** Use open and closed circles correctly.
[ ] Write a complete sentence explaining what each solution means in your case.

PART 4 - FIND THE MISTAKE: Error Analysis

Show that you can recognize and correct incorrect algebra.

[ ] Create one believable incorrect solution to an equation or inequality from your case.
[ ] Clearly mark the exact step where the mistake occurs.
[ ] Explain why that step is incorrect.
[ ] Redo the problem correctly and show all work.
[ ] Explain how the incorrect answer could lead to a poor real-world decision.

PART 5 - CASE CLOSED: Final Verdict

Use your evidence to solve the case.

[ ] Write a **5-7 sentence conclusion**.
[ ] Clearly state which option you recommend.
[ ] Use at least **3 pieces of mathematical evidence** from Parts 2 and 3.
[ ] Include evidence from at least one equation and one inequality.
[ ] Explain whether there are circumstances where the other option could be better.
[ ] End with: **Based on the mathematical evidence, I recommend...**

REQUIRED: RESOURCES USED

Add a final Canva page titled **Resources Used**. This page is required but is not one of the five mathematical parts.

For each source, include:

[ ] Name of the website, company, or organization
[ ] Title or description of the page you used
[ ] Web address (URL)
[ ] The information you used from that source
[ ] Date you accessed the information

SIMPLE SOURCE FORMAT

Example:
Company/Website: Example Wireless
Page Used: Unlimited Plans
Web Address: [full website address]
Information Used: Monthly plan price and data fee
Date Accessed: [date]

SOURCE EXPECTATIONS

Whenever possible, use the **official website** of the business or organization whose prices you are using. If you use another trustworthy source, make sure the information is current and clearly connected to your project. Do not copy large sections of website text; record only the facts you need and explain the project in your own words.

Your teacher should be able to use your Resources Used page to locate the information you researched.

POSSIBLE PROJECT IDEAS

Cell Phone Plans	Plan from Company A	Plan from Company B	Monthly fees, data costs, limits
School Dance Venues	Venue A	Venue B	Rental fees, per-person costs, capacity
Catering	Caterer A	Caterer B	Setup fees, price per person, minimum order
Gym Memberships	Gym A	Gym B	Monthly fees, joining fees, visit limits
Transportation	Service A	Service B	Starting fees, mileage rates
Printing	Printer A	Printer B	Setup fees, per-page prices, order minimums
Party Rentals	Rental A	Rental B	Base fee, guest charges, capacity
Savings Options	Savings Plan A	Savings Plan B	Starting amounts/rates or planned deposits

EXAMPLE: CELL PHONE PLAN INVESTIGATION

This is a teacher-created model using fictional prices. Your project must use your own Internet research and current information.

1. CASE FILE

Scenario: A student is comparing two fictional cell phone plans.
Plan A: $40 monthly fee + $5 per GB.
Plan B: $70 monthly fee + $2 per GB.
x = GB used per month.
Restriction: Monthly budget is $100 and expected use is 4-18 GB.
Central Question: Which plan should the student choose?

2. EVIDENCE #1 - EQUATIONS

Multi-Step: 40 + 5x = 90 -> 5x = 50 -> **x = 10 GB**. Check: 40 + 5(10) = 90. Plan A costs $90 at 10 GB.

Variables on Both Sides: 40 + 5x = 70 + 2x -> 3x = 30 -> **x = 10 GB**. At 10 GB, both plans cost $90.

3. EVIDENCE #2 - INEQUALITIES

Multi-Step: 40 + 5x <= 100 -> 5x <= 60 -> **x <= 12**. Plan A fits the $100 budget at 12 GB or less. Number line: closed circle at 12, shade left.

Compound: **4 <= x <= 18**. Expected use is at least 4 GB and no more than 18 GB. Number line: closed circles at 4 and 18, shade between.

4. FIND THE MISTAKE

Incorrect: 40 + 5x <= 100 -> 5x <= 140 -> x <= 28.
Mistake: 40 was added instead of subtracted. **Correct:** 40 + 5x <= 100 -> 5x <= 60 -> **x <= 12**. The wrong answer would make the student believe Plan A fits the budget at much higher data use.

5. CASE CLOSED

The plans cost the same at 10 GB. Plan A is less expensive below 10 GB, while Plan B becomes less expensive above 10 GB. Plan A fits the $100 budget only through 12 GB. The student's expected use may reach 18 GB, so Plan A may not always fit the budget. The student should compare expected data use with the researched prices before choosing. **Based on the mathematical evidence, Plan B is the better choice when usage is above 10 GB, as long as its total cost meets the student's budget.**

SUGGESTED CANVA ORGANIZATION

You may design your project creatively. This is a suggested page order that keeps the investigation easy to follow.

1 - Cover	Case title, your name, class/period, detective theme
2 - Case File	Scenario, Option A, Option B, researched costs, variables, restriction, central question
3 - Evidence #1	Two equations, all work, check, and explanations
4 - Evidence #2	Two inequalities, all work, number lines, and explanations
5 - Find the Mistake	Incorrect work, error explanation, corrected work, effect of mistake
6 - Case Closed	5-7 sentence final verdict with at least 3 pieces of mathematical evidence
7 - Resources Used	All websites/sources used for research

FINAL CHECKLIST

[ ] My Canva project has all 5 required parts and a Resources Used page.

[ ] I researched accurate/current costs on the Internet.

[ ] I did not use AI to complete any part of the project.

[ ] I used my own words and completed my own mathematics.

[ ] I included two equations with complete work and explanations.

[ ] I included two inequalities with complete work, number lines, and explanations.

[ ] My error analysis identifies, explains, and corrects a realistic mistake.

[ ] My 5-7 sentence verdict uses at least 3 pieces of mathematical evidence.

[ ] I listed every research source, its web address, information used, and access date.

[ ] My Canva pages are organized, readable, and visually clear.

CASE FILE: ALGEBRA

Equations & Inequalities Detective Investigation

ALGEBRA I HONORS - 7TH GRADE

FINAL PRODUCT	Canva Digital Project
PROJECT SCORE	100 Points
GRADE WEIGHT	70% of Major Assessment Grade
REQUIRED	5 Parts + Resources Used

PROJECT OVERVIEW

You are an **Algebra Detective**. Investigate a real-world situation in which someone must choose between two options. Research accurate and current costs on the Internet, use equations and inequalities as mathematical evidence, analyze an algebra mistake, and use your evidence to make a final recommendation.

Research Rule: Use reliable Internet sources, preferably official business or organization websites, to find accurate/current costs. Keep track of every source you use.

No-AI Rule: AI tools may not be used to choose the case, research prices, create or solve equations/inequalities, write explanations, write the verdict, or create the submitted Canva content/design. The submitted work must represent the student's own research, mathematics, writing, and design decisions.

Resources Used: Include a final page listing website/company, page used, web address, information used, and date accessed.

5 REQUIRED PARTS

1. CASE FILE - Scenario & Clues (15 pts)

Include a creative title; a clear real-world scenario; Option A and Option B; researched prices/fees; important clues and restrictions; variable definition with units; and a Central Question asking which option should be chosen.

2. EVIDENCE #1 - Solving Equations (25 pts)

Create and solve one multi-step equation and one equation with variables on both sides. Show every algebra step, check at least one solution by substitution, and explain in complete sentences what both answers mean in the real-world case.

3. EVIDENCE #2 - Solving Inequalities (25 pts)

Create and solve one multi-step inequality and one compound inequality. Show every algebra step, use inequality symbols correctly, graph both solutions accurately on number lines using correct open/closed circles and shading, and explain what both solutions mean.

4. FIND THE MISTAKE - Error Analysis (15 pts)

Create a believable incorrect equation or inequality solution. Identify the exact incorrect step, explain why it is wrong, redo the problem correctly, and explain how the incorrect result could affect the real-world decision.

5. CASE CLOSED - Final Verdict (15 pts)

Write a 5-7 sentence conclusion that clearly recommends one option. Use at least three pieces of mathematical evidence, including evidence from an equation and an inequality. Explain when the other option could be better and end with a clear evidence-based recommendation.

DETAILED GRADING RUBRIC

Use this rubric while you work. Each row tells you exactly how points are earned.

PART 1: CASE FILE			15
Scenario & Two Options	Scenario is realistic, clear, and detailed. Option A and Option B are easy to identify.	Scenario is present but unclear, too simple, or one option is not fully explained.	4
Research & Accurate Clues	Costs/fees and important facts are current, realistic, and based on Internet research.	Some information is missing, unrealistic, outdated, or not clearly researched.	4
Variable, Units & Restriction	Variable is clearly defined with units; case includes a meaningful budget, minimum, maximum, or range.	Variable/units or restriction is incomplete or unclear.	4
Central Question	Clear question asks which option should be chosen based on the evidence.	Question is vague or does not connect to both options.	3
PART 2: EVIDENCE #1 - EQUATIONS			25
Multi-Step Equation	Equation correctly models the case; all algebra steps are shown and correct.	Equation is connected to the case but contains modeling or solving errors.	8
Variables on Both Sides	Correct equation with variables on both sides; work is complete and accurate.	Equation is attempted but setup or solving contains errors.	8
Check & Interpretation	At least one solution is correctly checked; both answers are explained clearly with units and real-world meaning.	Check or explanations are missing, incomplete, or unclear.	5
Mathematical Communication	Work is organized, symbols are used correctly, and steps can be followed easily.	Work is difficult to follow or notation is inconsistent.	4
PART 3: EVIDENCE #2 - INEQUALITIES			25
Multi-Step Inequality	Correctly models a real restriction; all steps and final solution are accurate.	Model or solution contains errors but shows meaningful progress.	7
Compound Inequality	Correctly represents a meaningful range and is solved accurately.	Compound inequality is attempted but setup/solution is incomplete or incorrect.	7
Number Lines	Both solutions are graphed accurately with correct open/closed circles, direction, and shading.	One or both number lines contain errors or are incomplete.	5
Interpretation	Both inequality solutions are explained in complete sentences with real-world meaning and units.	One or both explanations are incomplete or do not connect to the case.	4
Notation & Work	Inequality notation and algebra steps are clear and mathematically correct.	Notation or steps contain errors that reduce clarity.	2
PART 4: FIND THE MISTAKE			15
Realistic Incorrect Work	Incorrect example is believable and connected to the student's case.	Error example is too obvious, unrelated, or incomplete.	3
Identify & Explain Error	Exact incorrect step is identified and the explanation clearly states why it is wrong.	Error is identified but explanation is weak or incomplete.	4
Corrected Work	Problem is redone correctly with complete algebra steps.	Correction is incomplete or contains additional errors.	4
Real-World Effect	Clearly explains how the mistake could change the decision or result.	Effect is mentioned but not clearly explained.	4
PART 5: CASE CLOSED			15
Final Recommendation	5-7 sentence conclusion clearly answers the Central Question and recommends an option.	Conclusion is too short/long, unclear, or recommendation is weak.	4
Use of Evidence	Uses at least 3 specific pieces of math evidence, including an equation and an inequality	Uses fewer than 3 pieces or evidence is vague/not mathematical	6

Reasoning & Other Option	Explains why the evidence supports the recommendation and when the other option might be better.	Reasoning is limited or does not consider the other option.	3
Clarity	Conclusion is written in complete sentences and is easy to understand.	Writing is difficult to follow or incomplete.	2
RESEARCH, RESOURCES & CANVA QUALITY			5
Resources Used	Final Resources Used page identifies sources, web addresses, information used, and access dates.	Sources are incomplete, difficult to trace, or missing required information.	3
Organization & Readability	Canva project is complete, organized, readable, and visually supports the math without distracting from it.	Project is difficult to read, disorganized, or missing required content.	2

STUDENT FINAL CHECKLIST

Check each item before submitting your Canva project.

[ ] I have all 5 required mathematical parts.

[ ] My case compares two clear real-world options.

[ ] I researched accurate/current costs on the Internet.

[ ] I defined my variable and included units.

[ ] I included a meaningful budget, limit, minimum, maximum, or range.

[ ] I solved one multi-step equation and one equation with variables on both sides.

[ ] I showed all equation work and checked at least one solution.

[ ] I explained what both equation answers mean.

[ ] I solved one multi-step inequality and one compound inequality.

[ ] I graphed both inequality solutions on number lines using correct open/closed circles and shading.

[ ] I explained what both inequality answers mean.

[ ] My Find the Mistake section shows incorrect work, identifies the exact error, explains it, and shows the correction.

[ ] My Case Closed section is 5-7 sentences and uses at least 3 pieces of mathematical evidence.

[ ] My Resources Used page lists every source, web address, information used, and access date.

[ ] I did not use AI to complete the project.

[ ] I checked my calculations, algebra, notation, units, spelling, and readability.

ACADEMIC INTEGRITY

By submitting this project, the student is stating that the research, mathematics, explanations, writing, and Canva content are the student's own work and that the project's no-AI requirement was followed.

Student Signature: _____ Date: _____

PROJECT PROPOSAL - GET YOUR CASE APPROVED

Complete this page before beginning your Canva project.

Student: _____ Period: _____

Case Title: _____

OPTION A: _____

OPTION B: _____

CENTRAL QUESTION: _____

1. Describe your real-world situation in 2-3 sentences:

_____

_____

2. What costs or information will you need to research online?

_____

_____

3. What limit or restriction will your case include?

_____

_____

4. Define your variable and unit:

_____

_____

5. List at least two websites or sources you plan to research:

_____

_____

Teacher Approval: _____ **Date:** _____

CASE STATUS: [ ] APPROVED [ ] REVISE & RESUBMIT

CANVA PROJECT DESIGN IDEA

PAGE 1

Example Case: Cell Phone Plan Investigation

CASE FILE:

CELL PHONE PLAN INVESTIGATION

Which plan is the better choice for me?

Student: _____

Class/Period: _____

Date: _____

PLAN A

$40 monthly fee

$5 per GB of data

PLAN B

$70 monthly fee

$2 per GB of data

Teacher-created Canva layout example - students should use their own research, math, writing, and design.

PART 1: CASE FILE

PAGE 2

Scenario & Clues

THE SCENARIO

I am comparing two cell phone plans. I researched current prices and fees from company websites. My goal is to choose the plan that fits my expected data use and monthly budget.

IMPORTANT CLUES

Plan A: $40 monthly fee + $5 per GB

Plan B: $70 monthly fee + $2 per GB

Monthly budget: no more than $100

Expected data use: 4 to 18 GB

VARIABLES

x = number of GB used in one month

C = total monthly cost in dollars

Always include units with your answers.

CENTRAL QUESTION

Which plan should I choose based on my expected data use and $100 monthly budget?

Teacher-created Canva layout example - students should use their own research, math, writing, and design.

PART 2: EVIDENCE #1

PAGE 3

Solving Equations

A. MULTI-STEP EQUATION

Question: When does Plan A cost $90?

$40 + 5x = 90$

$5x = 50$

$x = 10$ GB

CHECK:

$40 + 5(10) = 90$

$90 = 90$

Meaning: Plan A costs $90 when 10 GB of data are used.

B. VARIABLES ON BOTH SIDES

Question: When do both plans cost the same?

$40 + 5x = 70 + 2x$

$3x = 30$

$x = 10$ GB

At 10 GB:

Plan A = $90

Plan B = $90

Meaning: The plans have the same cost at 10 GB.

PART 3: EVIDENCE #2

PAGE 4

Solving Inequalities + Number Lines

A. MULTI-STEP INEQUALITY

Budget rule: Plan A must cost no more than $100.

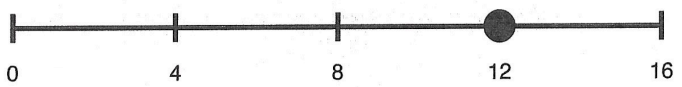
$40 + 5x <= 100$

$5x <= 60$

$x <= 12$

Meaning: Plan A stays within budget at 12 GB or less.

NUMBER LINE



Closed circle at 12; shade left.

B. COMPOUND INEQUALITY

Expected use is between 4 GB and 18 GB.

$4 <= x <= 18$

Meaning: Data use is at least 4 GB and no more than 18 GB.

NUMBER LINE



Teacher-created Canva layout example - students should use their own research, math, writing, and design.

PART 4: FIND THE MISTAKE

PAGE 5

Error Analysis

INCORRECT WORK

40 + 5x <= 100

5x <= 140

x <= 28

The mistake happens on the second line.

WHAT IS THE MISTAKE?

The student added 40 instead of subtracting 40 from both sides. To isolate 5x, subtract 40 from 100.

CORRECT WORK

40 + 5x <= 100

5x <= 60

x <= 12

WHY IT MATTERS

The incorrect answer would make the student believe Plan A fits the $100 budget up to 28 GB. That could lead to an unexpectedly high bill.

PART 5: CASE CLOSED

PAGE 6

Final Verdict

MY CONCLUSION

The plans cost the same at 10 GB. Plan A is less expensive below 10 GB, while Plan B becomes less expensive above 10 GB. Plan A stays within the $100 budget only through 12 GB. My expected data use may be as high as 18 GB, so Plan A may not always fit the budget. The inequality evidence shows that my budget is an important limit. I should compare my usual data use with the researched prices before making a final choice. Based on the mathematical evidence, I recommend Plan B when I expect to use more than 10 GB, as long as its total cost stays within my budget.

SUPPORTING EVIDENCE

1. Both plans = $90 at 10 GB.
2. Plan A budget limit: x <= 12.
3. Expected use: 4 <= x <= 18.

FINAL DECISION

Use your math to explain your recommendation. Mention when the other option could be better.

RESOURCES USED

PAGE 7

Required Final Canva Page

Website / Company	Page Used	Information Used	Web Address	Date
Example Wireless	Plans	Monthly fee + data cost	company website	10/15/26
Example Mobile	Plans	Monthly fee + data cost	company website	10/15/26

Use the Internet for accurate/current costs. List every source. Do not use AI to complete the project.

Teacher-created Canva layout example - students should use their own research, math, writing, and design.